Sales & Supply Chain Performance Analysis using SQL

Turning Data into Business Insights

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Business Context

- The company operates across multiple markets and product categories
- Handles large volumes of sales and forecast data
- Requires better visibility into business performance

Focus Areas

- Sales trends
- Customer contribution
- Supply chain efficiency

Problem Statement

- Forecasted demand often differs from actual sales
- Limited visibility into key revenue drivers
- Difficulty in identifying performance gaps across customers and markets

To analyze

- sales
- customer contribution
- forecast accuracy

to support better decision-making

Data Overview

Fact Tables

- fact_sales_monthly → Contains sales transactions data
- fact_act_est → Contains forecast vs actual data

Dimension Tables

- dim_customer → Customer details (market, region, channel)
- dim_product → Product details (category, division)

Data Includes

- Sales Quantity & Revenue
- Forecast Quantity
- Customer & Product Information

Tools & Skills Used

Tools

- (MySQL Workbench) for data analysis

Skills

- Joins to combine multiple tables
- Aggregations for KPI calculations
- CTEs (Common Table Expressions) for structured queries
- Window Functions for ranking and contribution analysis

Market Presence Analysis

- Filtered data for "Atliq Exclusive" in APAC region
  - Focused on identifying operating markets

```
SELECT market FROM gdb0041.dim_customer
where customer = "Atliq Exclusive" and region = "APAC"
group by market
order by market;
```

INPUT QUERY

market
Australia
Bangladesh
India
Indonesia
Japan
Newzealand
Philiphines
South Korea

OUTPUT

Year-over-Year Product Growth Analysis

- Compared unique product count between 2020 and 2021
- Used aggregation to measure growth in product offerings

```sql
WITH X AS
(SELECT COUNT(DISTINCT product_code) AS unique_products_2020
FROM  fact_sales_monthly  WHERE fiscal_year= 2020),
Y AS
(SELECT COUNT(DISTINCT product_code) AS unique_products_2021
FROM fact_sales_monthly WHERE fiscal_year= 2021)
SELECT
X.unique_products_2020,
Y.unique_products_2021,
round(((Y.unique_products_2021-X.unique_products_2020)/X.unique_products_2020)*100,2)
AS Percentage_chg FROM X,Y;
```

INPUT QUERY

unique_products_2020	unique_products_2021	Percentage_chg
245	334	36.33

OUTPUT

Product Distribution Across Segments

- Analyzed number of products in each segment
- Used GROUP BY to understand product distribution

```sql
select * from dim_product;
select
        segment,
        count(distinct(product_code)) as product_count
from dim_product
group by segment
order by product_count desc;
```

INPUT QUERY

segment	product_count
Notebook	129
Accessories	116
Peripherals	84
Desktop	32
Storage	27
Networking	9

OUTPUT

Segment-wise Product Growth Analysis

- Compared product count by segment for 2020 and 2021
  - Identified which segment had the highest growth

```sql
with cte1 as(
select  dp.segment as A, count(distinct fs.product_code) as B
from fact_sales_monthly fs
join dim_product dp
on fs.product_code=dp. product_code
group by dp.segment , fs.fiscal_year
having fs.fiscal_year=2020
),
 cte2 as(
select  dp.segment as C, count(distinct fs.product_code) as D
from fact_sales_monthly fs
join dim_product dp
on fs.product_code=dp. product_code
group by dp.segment, fs.fiscal_year
having fs.fiscal_year= 2021
)
select cte1.A as segment,
        cte1.B as product_code_2020,
        cte2. D as product_code_2021,
        (cte2.D-cte1.B) as difference
from cte1,cte2
where cte1.A=cte2.C;
```

INPUT QUERY

segment	product_code_2020	product_code_2021	difference
Accessories	69	103	34
Desktop	7	22	15
Networking	6	9	3
Notebook	92	108	16
Peripherals	59	75	16
Storage	12	17	5

OUTPUT

Products with Highest & Lowest Manufacturing Cost

- Identified products with maximum and minimum manufacturing cost
- Used subqueries to extract extreme cost values

```sql
select
        m.product_code,
        p.product,
        m.manufacturing_cost
from fact_manufacturing_cost m
join dim_product p
on m.product_code= p.product_code
where manufacturing_cost in
(
select max(manufacturing_cost) from fact_manufacturing_cost
union
select min(manufacturing_cost) from fact_manufacturing_cost
)
order by manufacturing_cost desc;
```

INPUT QUERY

product_code	product	manufacturing_cost
A6121110208	AQ HOME Allin1 Gen 2	263.4207
A2118150101	AQ Master wired x1 Ms	0.8654

OUTPUT

Top 5 Customers by Average Discount (India, 2021)

- Identified customers with highest average pre-invoice discount
- Focused on Indian market for FY 2021
- Used CTE + aggregation + filtering for analysis

```sql
with cte1 as (select customer_code as A , Avg(pre_invoice_discount_pct) as B
 from fact_pre_invoice_deductions
 where fiscal_year = 2021
 group by customer_code),

cte2 as ( select customer_code as C, customer as  D
 from dim_customer
 where market ="India")

select cte2.C as customer_code,
        cte2.D as cutsomer,
        round(cte1.B,4) as Average_discount_percentage
from cte1,cte2
where cte1.A= cte2.C
order by Average_discount_percentage desc
limit 5;
```

INPUT QUERY

customer_code	cutsomer	Average_discount_percentage
90002009	Flipkart	0.3083
90002006	Viveks	0.3038
90002003	Ezone	0.3028
90002002	Croma	0.3025
90002016	Amazon	0.2933

OUTPUT

Monthly Gross Sales Analysis (Atliq Exclusive)

- Analyzed monthly gross sales performance for Atliq Exclusive
- Combined data from sales, customer, and pricing tables using joins

```
select monthname(s.date)as month,s.fiscal_year,
round(sum(g.gross_price*sold_quantity),2)
as gross_sales_amt from fact_sales_monthly s
join dim_customer c using(customer_code)
join fact_gross_price g using(product_code)
where customer="atliq exclusive"
group by monthname(s.date),s.fiscal_year
order by fiscal_year;
```

INPUT QUERY

month	fiscal_year	gross_sales_amt
September	2018	2347703.88
October	2018	2462780.55
November	2018	3766114.43
December	2018	2390015.56
January	2018	2285937.67
February	2018	1985466.36
March	2018	2219880.14
April	2018	1392024.51
May	2018	2310946.52
June	2018	1976109.61
July	2018	2224693.76
August	2018	1498728.56
September	2019	7860039.25

OUTPUT

Quarter-wise Sales Performance (2020)

- Converted monthly data into quarter-wise sales using CASE statement
- Analyzed total sold quantity per quarter

```sql
SELECT
    CASE
        WHEN MONTH(date) IN (9,10,11) THEN 'Q1'
        WHEN MONTH(date) IN (12,01,02) THEN 'Q2'
        WHEN MONTH(date) IN (03,04,05) THEN 'Q3'
        ELSE 'Q4'
    END AS Quarters,
    SUM(sold_quantity) AS total_sold_quantity
FROM fact_sales_monthly
WHERE fiscal_year = 2020
GROUP BY Quarters
ORDER BY total_sold_quantity DESC;
```

INPUT QUERY

Quarters	total_sold_quantity
Q1	7005619
Q2	6649642
Q4	5042541
Q3	2075087

OUTPUT

Sales Contribution by Channel (2021)

- Analyzed gross sales contribution by different channels
- Calculated percentage contribution of each channel

```
with x as(select c.channel,
round(sum(g.gross_price*s.sold_quantity)/100000,2)as gross_sales_mln
from fact_sales_monthly s
join dim_customer c using(customer_code)
join fact_gross_price g using(product_code)
group by c.channel)

select  channel, gross_sales_mln,
round((gross_sales_mln/(select sum(gross_sales_mln)from x))*100,2)
as pct from x
order by gross_sales_mln desc
```

INPUT QUERY

channel	gross_sales_mln	pct
Retailer	124419.42	72.70
Direct	26742.67	15.63
Distributor	19986.24	11.68

OUTPUT

Top 3 Products by Division (2021)

- Identified top-performing products within each division
- Used window function (RANK) to rank products based on sales

```sql
WITH X AS ( SELECT
    P.division, S.product_code, P.product,
    SUM(S.sold_quantity) AS total_sold_quantity,
RANK() OVER(PARTITION BY P.division
ORDER BY SUM(S.sold_quantity) DESC) AS Rank_Order
FROM dim_product P
JOIN fact_sales_monthly S ON P.product_code = S.product_code
WHERE S.fiscal_year = 2021
GROUP BY P.division, S.product_code, P.product)

SELECT division, product_code, product,
    total_sold_quantity, Rank_Order
FROM X
WHERE Rank_Order IN (1,2,3)
ORDER BY division, Rank_Order;
```

INPUT QUERY

division	product_code	product	total_sold_quantity	Rank_Order
N & S	A6720160103	AQ Pen Drive 2 IN 1	701373	1
N & S	A6818160202	AQ Pen Drive DRC	688003	2
N & S	A6819160203	AQ Pen Drive DRC	676245	3
P & A	A2319150302	AQ Gamers Ms	428498	1
P & A	A2520150501	AQ Maxima Ms	419865	2
P & A	A2520150504	AQ Maxima Ms	419471	3
PC	A4218110202	AQ Digit	17434	1
PC	A4319110306	AQ Velocity	17280	2
PC	A4218110208	AQ Digit	17275	3

OUTPUT

Key Insights & Business Impact

Key Insights

- Product portfolio expanded by 36%+ YoY, indicating strong business growth
- Certain segments dominate product distribution, showing focused investment areas
- Top customers contribute a major share of revenue, highlighting dependency on key accounts
- Retailer channel contributes ~70%+ of total sales, making it the primary revenue driver
- Seasonal trends observed in monthly & quarterly analysis, helping in demand forecasting
- Significant variation in manufacturing costs, indicating differences in product complexity
- Top products within each division drive maximum sales, supporting targeted product strategy

Final Business Impact

These insights enable data-driven decisions in product strategy, customer targeting, channel optimization, and cost management.

THANK YOU